MILITARY SPECIFICATION

MONITORS, MOTOR INSULATION RESISTANCE, SHIPBOARD USE (METRIC)

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

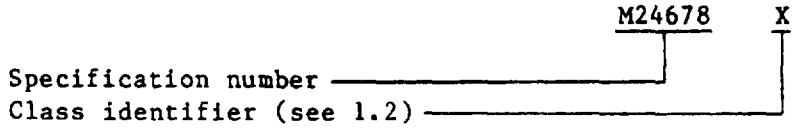
1. SCOPE

1.1 Scope. This specification covers the requirements for devices that continuously monitor the insulation resistance to ground of de-energized alternating current (ac) and direct current (dc) motors. Monitors provide local and remote indications that a specific minimum value of insulation resistance has been measured. These monitors are for installation in motor controllers and related electrical equipment for shipboard use. The monitors are also for use with motors driving auxiliary equipment. The monitors are not for use with generators, with motors driving generators or with propulsion motors.

1.2 Classification. Monitors shall be of the following classes as specified (see 3.3.1.1, 3.3.1.2 and 6.2.1):

Class 1 - Ac motor monitors
Class 2 - Dc motor monitors

1.3 Part numbers. Part numbers under this specification shall be formulated as follows (see 6.2.1).



Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-F-320 - Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

MILITARY

MIL-S-901 - Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-E-917 - Electric Power Equipment, Basic Requirements (Naval Shipboard Use).
MIL-E-2036 - Enclosures for Electric and Electronic Equipment, Naval Shipboard.
MIL-C-2212 - Controllers, Electric Motor A.C. or D.C., and Associated Switching Devices.
MIL-P-15024 - Plates, Tags and Bands for Identification of Equipment.
MIL-P-15024/5 - Plates, Identification.
MIL-E-17555 - Electronic and Electrical Equipment, Accessories and Provisioned Items (Repair Parts): Packaging of.
MIL-L-19140 - Lumber and Plywood, Fire Retardant Treated.
MIL-T-55164 - Terminal Boards, Molded, Barrier, Screw and Stud Types, and Associated Accessories, General Specification for.
MIL-T-55164/6 - Terminal Boards, Molded, Barrier, Screw Type, Class 42 TB.

STANDARDS

MILITARY

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-454 - General Requirements for Electronic Equipment.
MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-965 - Parts Control Program.

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MILITARY (Continued)

DOD-STD-1399, Section 300 - Interface Standard for Shipboard Systems, Electric Power, Alternating Current. (Metric)
DOD-STD-2142 - Magnetic Silencing Characteristics, Measurement of. (Metric)
DOD-STD-2143 - Magnetic Silencing Requirements for the Construction of Nonmagnetic Ships and Craft. (Metric)

2.1.2 Other Government publication. The following other Government publication forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

PUBLICATION

TE-000-AB-GTP-010 - Parts Application and Reliability Information Manual for Navy Electronic Equipment.

(Application for copies should be addressed to the Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

(Copies of specifications, standards, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Materials, parts and processes. Unless otherwise specified, materials, parts and processes shall be in accordance with MIL-E-917. Monitors shall be designed to the parts derating criteria of NAVSEA TE-000-AB-GTP-010. Parts which operate at stresses exceeding the specified derating criteria shall not be used prior to approval by the contracting activity. At maximum environmental stress, electrical and electronic parts shall not be thermally or electrically stressed greater than the maximum ambient temperatures or maximum percentages of rated electrical stress listed in table I of NAVSEA TE-000-AB-GTP-010. The contractor shall establish a parts control program in accordance with MIL-STD-965.

3.2.1 Recovered materials. Unless otherwise specified herein, all equipment and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended

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use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification.

3.2.2 Hazardous items and toxic materials. The material identified in MIL-E-917 as prohibited material shall not be used in the construction of the monitors. Questions pertinent to this effect shall be referred by the contracting activity to the appropriate departmental medical service who will act as an advisor to the contracting activity. Regardless of any other requirements, materials and parts containing asbestos, cadmium, lithium, mercury, radioactive materials or materials giving off toxic fumes under operating or casualty conditions shall not be used.

3.2.3 Material compatibility. Materials which, through outgassing or other physical phenomena, cause deterioration of any other materials or degradation of equipment performance shall not be used.

3.3 Performance.

3.3.1 Input power. With rated input voltage applied, monitors shall require no more than 10 watts to perform their operation. When more than one input voltage is required for the operation of an individual monitor, all input voltages shall be the same magnitude, frequency and phase (see 4.7.2).

3.3.1.1 Class 1 monitors. Class 1 monitors shall meet all requirements specified herein when supplied with type I or type II power as defined in DOD-STD-1399, section 300 except that monitors shall meet all requirements specified herein with a continuous average variation of plus or minus 20 percent of nominal input voltage and a continuous average variation from rated frequency of plus or minus 4 percent of rated frequency. Nominal input voltage shall be either 440 volts alternating current (Vac) or 115 Vac single phase power as specified (see 4.7.2 and 6.2.1).

3.3.1.2 Class 2 monitors. Class 2 monitors shall meet all the requirements specified herein when supplied with the nominal input voltage of table I as specified (see 6.2.1), and over the voltage range specified in table I and voltage spike specified in 4.7.13 without any change in operation (see 4.7.2).

TABLE I. Voltage and voltage range for class 2 monitors.

Nominal input voltage (volts direct current (Vdc))	Voltage range (Vdc)
115	92 to 127
250	180 to 355
500	350 to 710

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3.3.2 Operation. Monitors shall be energized to monitor motor insulation resistance only when power is removed from the motor being monitored. Monitors shall be isolated from connection to motor windings while motors are energized. Monitors shall not prevent starting or running of motors. When monitors are energized, monitors shall measure the winding leakage current to ground in order to determine the value of insulation resistance of electric motor windings. Monitors shall be connected to motor windings using isolation relays located within the enclosure of the monitor (see 3.4.6). When energized, monitors shall apply a potential of no greater than 24 Vdc between motor windings and ground. Monitors shall have current limit protection that limits the current applied by the monitor to the motor windings to a maximum value of 0.0001 Adc under any condition of operation. A monitor shall provide an indication when the insulation resistance of motor windings is 1 megohm (plus 5 percent) or less by energizing a relay located within the enclosure of the monitor (see 3.4.6). The relay shall energize indicator lights that are in accordance with MIL-C-2212. One light for local indication of low insulation resistance shall be provided with each monitor in order to mount on the motor controller enclosure containing the monitor. One light for remote indication of low insulation resistance shall be provided for each start/stop station of the motor being monitored (see 4.7.3).

3.3.3 Inclined operation. Monitors shall meet all requirements specified herein when permanently inclined at any angle up to 60 degrees in any direction from the normal operating position (see 4.7.3).

3.3.4 Duty. Monitors shall operate continuously whenever power is applied to the monitors.

3.3.5 Endurance. Monitors shall operate for a minimum of 100,000 cycles without failure of any component. One cycle is defined as the application and removal of input power to the monitor (see 4.7.14).

3.4 Construction. Unless otherwise specified, monitors and remote indicators shall be in accordance with MIL-E-917. When specified (see 6.2.1), monitors and remote indicators for nonmagnetic applications shall be in accordance with DOD-STD-2143 (see 4.7.12).

3.4.1 Enclosure. Components of the monitor, with the exception of the remote indicator, shall be contained in one enclosure. Enclosures for monitors and remote indicators shall be in accordance with MIL-E-2036 for class 1, totally enclosed equipment. Mounting brackets shall be permanently attached to the enclosure. Each enclosure shall be of adequate strength to support itself and its components and meet all the requirements specified herein when mounted (see 4.6).

3.4.2 Painting. Parts not fabricated of a corrosion-resisting material or having a corrosion-resistant treatment as specified in MIL-E-917 shall be painted as specified therein. Touching-up is permitted for marks or scratches due to assembly, testing, or other handling (see 4.6).

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3.4.3 Mounting. Monitors and remote indicators shall meet the mounting requirements in accordance with MIL-E-917. Monitors shall mount solidly inside motor controllers in accordance with MIL-C-2212. Enclosures for remote indicators shall mount solidly adjacent to each remote start/stop station. Mounting bolts shall be readily accessible from outside the enclosure of the monitor or the enclosure of the remote indicators (see 4.6).

3.4.4 Electrical connection. Electrical connection to monitors shall be in accordance with MIL-C-2212. Electrical connection to monitors and remote indicators shall be made using terminal boards in accordance with MIL-T-55164 and MIL-T-55164/6, screw type, class 42TB. Terminal boards shall be labelled in accordance with MIL-E-917 (see 4.6).

3.4.5 Electrical creepage and clearance distances. Creepage and clearance distances shall be in accordance with MIL-E-917 (see 4.6).

3.4.6 Relays. Relays used in the operation of monitors shall be located within the enclosure of the monitor and shall be rated for continuous duty. Relay contacts of isolation relays that connect the monitor to the motor winding shall not chatter (change state) in either the energized or de-energized position or during transition of position or during shock (see 4.7.7). Relay contacts of relays used in the operation of indicator lights shall not permanently change state during shock (see 4.7.7) or change state during vibration (see 4.7.8).

3.4.7 Dimensions and weight. Monitors, including mounting brackets and enclosures, shall not exceed the dimensions and weights specified in table II (see 4.7.1).

TABLE II. Dimensions and weights.

Class	Height-inches (mm)	Width-inches (mm)	Depth-inches (mm)	Weight-pounds (kg)
1	3.0 (76)	2.5 (63.5)	5.0 (127)	2.2 (1.0)
2	3.0 (76)	4.75 (120)	5.0 (127)	2.2 (1.0)

3.4.8 Electrical isolation. The electrical isolation between the input power to the monitor and the monitor's connections to motor windings, control voltages and any circuits, relays and relay contacts of the monitor that are used to connect the monitor to the motor windings, control voltages or remote indicators shall be not less than 100 megohms at 77 degrees Fahrenheit (°F) (25 degrees Celsius (°C)) when tested as specified in 4.7.4.

3.4.9 Insulation resistance. The insulation resistance between any electrical connection to the monitor and ground shall be not less than 100 megohms at 77°F (25°C) when tested as specified in 4.7.5.

3.4.10 Dielectric withstanding voltage. Monitors shall meet the dielectric withstanding voltage test requirements of 4.7.6.

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3.4.11 Maintainability. The lowest level of repair of monitors shall be the monitor and the remote indicator. Maintainability of monitors shall be by the replacement of the monitor or remote indicator as required.

3.5 Environment requirements.

3.5.1 Shock. Monitors shall conform to shock requirements for grade A, class 1, type A lightweight equipment in accordance with MIL-S-901.

3.5.2 Vibration. Monitors shall withstand type I vibration tests up to and including 33 hertz (Hz) in accordance with MIL-STD-167-1.

3.5.3 Electromagnetic interference (EMI) emission and susceptibility. Monitors shall conform to the EMI limits in accordance with MIL-STD-461 for the requirements of CE01, CE03, CS01, RE01, RE02 and RS01.

3.5.4 Ambient temperature. Monitors shall start and operate as specified herein throughout an ambient temperature range of 32 to 149°F (0 to 65°C).

3.5.5 Humidity. Monitors shall operate as specified herein when tested as specified in 4.7.11.

3.6 Identification marking. Each monitor shall be provided with an identification plate. Identification plates shall conform to type A, B, or C in accordance with MIL-P-15024 and MIL-P-15024/5. Identification plates shall be securely fastened to the monitors. Identification plates shall contain the following information:

(a) Manufacturer's name and model number
(b) National stock number
(c) Class
(d) Rated voltage
(e) Wiring diagram detailing all external connections to the terminal board of the monitor.

3.7 Drawings. When specified in the contract or order, drawings shall be prepared (see 6.2.2).

3.7.1 Certificate of compliance. When specified in the contract or order a certificate of compliance shall be prepared (see 6.2.2).

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3.8 Workmanship. Monitors, including all components, parts and accessories, shall exhibit neatness and thoroughness of soldering, marking, wiring, welding, brazing, plating, riveting, finishing and freedom of burrs and sharp edges. Workmanship shall conform to requirement 9 of MIL-STD-454. Soldering shall conform to requirement 5 of MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.4).
(b) Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in MIL-STD-202. Test conditions of temperature and humidity shall be measured.

4.4 First article inspection. First article inspection shall consist of the examination and tests specified in table III for groups A, B, C and D. First article inspection shall be performed on assembled monitors.

4.4.1 Sequence of tests. The operation test shall be conducted at the conclusion of the dielectric withstanding voltage test and all group C tests. Otherwise, the sequence of tests may be determined by the contractor.

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TABLE III. First article and quality conformance inspections.

Examination or test	Requirement	Examination or test	First article	Quality 1/ conformance
Group A				
Examination	3.2, 3.4.1 through 3.4.5, 3.5.5, 3.6, 3.8	4.6	X	X
Size and weight	3.4.7	4.7.1	X	X
Group B				
Input power and voltage tolerance	3.3.1, 3.3.1.1, 3.3.1.2	4.7.2	X	X
Operation and output power	3.3.2 and 3.3.3	4.7.3	X	X
Electrical isolation	3.4.8	4.7.4	X	X
Insulation resistance	3.4.9	4.7.5	X	X
Dielectric withstanding voltage	3.4.10	4.7.6	X	X
Group C				
Shock	3.5.1	4.7.7	X	
Vibration	3.5.2	4.7.8	X	
EMI emission and susceptibility	3.5.3	4.7.9	X	
Ambient temperature	3.5.4	4.7.10	X	
Humidity	3.5.5	4.7.11	X	
Magnetic characteristics (for nonmagnetic applications only)	3.4	4.7.12	X	
Voltage spike	3.3.1.1 and 3.3.1.2	4.7.13	X	
Group D				
Endurance	3.3.5	4.7.14	X	

1/ Inspections for each monitor. For periodicity of quality conformance inspections for groups C and D (see 4.5).

4.4.2 Sample. One monitor of each class shall be subjected to first article inspection. For any design change, one monitor shall be subjected to first article inspection. The monitor shall be taken at random from a production run and shall be produced with equipment and procedures normally used in production.

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4.4.3 Failures. Failure of any monitor to pass any first article inspection test for that class monitor shall be cause for rejection of any lot offered for delivery. Monitors shall be accepted for delivery only after the cause of failure has been identified, the repair demonstrated by first article inspection test and the deficiency corrected on all monitors offered for delivery and all future monitors.

4.4.4 First article inspection report. When specified in the contract or order, a first article inspection report shall be prepared (see 6.2.2).

4.5 Quality conformance inspection. Each monitor shall be subjected to and pass the quality conformance inspections for groups A and B examinations or tests specified in table III. One monitor of each class for every 150 monitors of each class shall be subjected to and pass periodic inspection for groups C and D tests in addition to groups A and B examinations and tests specified in table III.

4.5.1 Noncompliance. If a sample fails to pass group C or D inspection, the contractor shall notify the contracting activity of such failure and take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which are manufactured under essentially the same materials and processes, and which are considered subject to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action, acceptable to the contracting activity has been taken. After the corrective action has been taken, groups C and D inspection shall be repeated on additional sample units (all tests and examinations, or the test which the original sample failed, at the option of the contracting activity). Groups A and B inspections may be reinstituted; however, final acceptance and shipment shall be withheld until the group C or D inspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure shall be furnished to the cognizant inspection activity and the contracting activity.

4.5.2 Quality conformance inspection report. When specified in the contract or order, a quality conformance inspection report shall be prepared (see 6.2.2).

4.6 Examinations. Monitors shall be examined to determine that the material, workmanship, safety to personnel, design and construction are in conformance with section 3 and the following:

(a) Materials are in accordance with MIL-E-917 (see 3.2).
(b) Enclosures are in accordance with MIL-E-2036 (see 3.4.1).
(c) Painting is in accordance with MIL-E-917 (see 3.4.2).
(d) Mounting provisions are in accordance with MIL-E-917 (see 3.4.3).
(e) Terminal boards and markings are in accordance with MIL-E-917 (see 3.4.4).
(f) Creepage and clearance distances are in accordance with MIL-E-917 (see 3.4.5).
(g) Identification plates and marking are in accordance with MIL-P-15024, MIL-P-15024/5 and 3.6.
(h) Workmanship is in accordance with 3.8.

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4.7 Tests.

4.7.1 Size and weight test. The size and weight of monitors shall be measured and recorded to determine conformance with 3.4.7. Dimensions and weights of monitors exceeding those specified in table II shall constitute a failure.

4.7.2 Input power and voltage tolerance test.

4.7.2.1 Input power test. Monitors shall be connected to an input voltage source which shall supply nominal input voltage at nominal frequency as specified in 3.3.1.1 for class 1 monitors and 3.3.1.2 for class 2 monitors. Monitors shall be connected in a manner simulating a shipboard installation and operated in a normal manner that maximizes input power to the monitor. Measurements of input voltage, input current and input power in watts shall be recorded with nominal input voltage applied to the monitor. Input power greater than that specified in 3.3.1 shall constitute a failure.

4.7.2.2 Voltage tolerance test. Monitors shall be connected to an input voltage source which shall be varied in voltage (plus or minus 20 percent) and frequency (plus or minus 4 percent) and harmonic voltage content (5 percent total voltage harmonic distortion, 3 percent maximum single voltage harmonic) for class 1 monitors as specified in 3.3.1.1 and varied in voltage for class 2 monitors as specified in 3.3.1.2. Monitors shall be connected to a continuously adjustable resistance load that shall be varied to cause the monitor to measure a resistance simulating insulation resistance from 0.5 to 1.5 megohms. For class 1 monitors, the following input power conditions shall be tested.

(a) Minimum voltage and minimum frequency.
(b) Minimum voltage and maximum frequency.
(c) Nominal voltage and nominal frequency.
(d) Maximum voltage and minimum frequency.
(e) Maximum voltage and maximum frequency.

For class 2 monitors, the minimum, nominal and maximum supply voltages shall be tested. With the conditions of input power established, the value of insulation resistance shall be varied from 1.5 to 0.5 megohm and from 0.5 to 1.5 megohms. The value of resistance for each condition of input power shall be measured and recorded whenever the monitor indicates that a change in the insulation resistance has been sensed.

4.7.2.3 Definition of failure. The following are definitions of failure to conform to the voltage tolerance requirements specified in 4.7.2.2:

(a) Failure to sense and indicate that resistance has decreased to 1.0 megohm within a range of plus or minus 5 percent.
(b) Failure to continue indicating that resistance is less than 1.0 megohm.
(c) Failure to sense and cease indicating that resistance is greater than 1.05 megohms.

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4.7.3 Operation. Monitors shall be connected to an input voltage source that shall supply nominal input voltage (plus or minus 20 percent) at nominal frequency (plus or minus 4 percent) as specified in 3.3.1.1 for class 1 monitors and minimum, nominal and maximum supply voltages as specified in 3.3.1.2 for class 2 monitors. Monitors shall be connected to an integral horsepower electric motor and motor controller in a manner simulating a shipboard installation. Motor insulation resistance shall be at least 10 megohms at 77°F (25°C). For class 1 monitors, the following input power conditions shall be tested:

(a) Minimum voltage and minimum frequency.
(b) Minimum voltage and maximum frequency.
(c) Nominal voltage and nominal frequency.
(d) Maximum voltage and minimum frequency.
(e) Maximum voltage and maximum frequency.

For class 2 monitors, the minimum, nominal and maximum supply voltages shall be tested. The motor shall be energized to maximum speed. Grounding of the motor winding during acceleration or at maximum speed by the monitor shall constitute a failure of the monitor. The motor shall be de-energized. Failure to sense acceptable motor insulation resistance after the motor has come to rest shall constitute a failure of the monitor. A continuously adjustable resistor adjustable from 0.5 to 1.5 megohms shall be connected between a stator coil and ground. The resistance shall be decreased from 1.5 megohms until the monitor senses and indicates that a low value of insulation resistance exists. The monitor shall be disconnected from the motor and insulation resistance shall be measured with a calibrated insulation resistance-measuring instrument. A value of insulation resistance less than 0.95 megohm or greater than 1.05 megohms shall constitute a failure. The test shall be repeated for each power condition with the monitor inclined 60 degrees in each direction from each of the three principal axes of the monitor. The same failure criteria shall apply.

4.7.3.1 Output power test. Monitors shall be connected to an input voltage source that shall supply the maximum supply voltage at nominal frequency as specified in 3.3.1.1 for class 1 monitors and the maximum supply voltage as specified in 3.3.1.2 for class 2 monitors. With a voltmeter connected between the monitor's output connections to the motor and ground, the monitor shall be energized. A voltage greater than 24 Vdc shall constitute a failure of the monitor. With the monitor's output connections to the motor and ground short circuited, the monitor shall be energized. An output current greater than 0.0001 Adc shall constitute a failure of the monitor.

4.7.4 Electrical isolation tests. Electrical isolation tests shall be conducted in accordance with method 302 of MIL-STD-202 and shall conform to the requirements of 3.4.8. Test conditions shall verify input/output isolation and shall be as follows:

(a) Test potential: Test condition B for monitors with rated supply voltage less than 500 volts. Test condition C for monitors with rated supply voltage greater than or equal to 500 volts.

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(b) Points of measurement: Between each power input connection to the monitor and each output connection of the monitor to motor windings, control voltages and remote indicators.
(c) Electrification time: 60 seconds minimum.

4.7.5 Insulation resistance tests. Insulation resistance tests shall be conducted in accordance with method 302 of MIL-STD-202 and shall conform to the requirements of 3.4.9. Test conditions shall be as follows:

(a) Test potential: Test condition B for monitors with rated supply voltage less than 500 volts. Test condition C for monitors with rated supply voltage greater than or equal to 500 volts.
(b) Points of measurement: Between each electrically isolated circuit and all other circuits and ground (see 4.7.5.1).
(c) Electrification time: 60 seconds minimum.

4.7.5.1 Electrically isolated circuits shall be determined by application of the following criteria:

(a) Circuits, whose only connection to each other is by electromagnetic coupling through a magnetic core which is shared in common by the circuits, shall be considered to be electrically isolated from each other.
(b) Circuits, whose only connection to each other is through a capacitor, shall not be considered to be electrically isolated from each other. When the purpose is to test circuits internal to the equipment, such circuits shall be temporarily interconnected with a jumper wire or test lead.

4.7.6 Dielectric withstanding voltage tests. Dielectric withstanding voltage tests shall be conducted in accordance with method 301 of MIL-STD-202. Test conditions shall be as follows:

(a) Preparation: The monitor shall be mounted to a test fixture in accordance with required installation directions.
(b) Test voltage: The root mean square (rms) ac test voltage shall be twice rated input voltage plus 1000 Vac (rms).
(c) Points of application: The test voltage shall be applied between electrically isolated circuits of the monitor and between all electrical circuits and ground.
(d) Duration of test: The duration of the test shall be 60 seconds minimum.
(e) Definition of failure: Any arcing or disruptive discharge as evidenced by instantaneous change in nominal leakage-current or failure of the monitor to pass the operation test of 4.7.3 shall constitute a failure.

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4.7.7 Shock test. Monitors shall be subjected to a shock test to determine conformance with 3.5.1 for grade A, class 1, type A lightweight equipment. Monitors shall be mounted to the mounting plate using fixture 6E of MIL-S-901. Monitors shall be mounted on the shock machine in a manner simulating normal shipboard mounting. During the conduct of the shock test, monitors shall alternate state of operation for each blow starting with being energized and in operation. The test procedure for monitors shall be as specified for lightweight equipment in accordance with MIL-S-901. The state of all outputs of the monitor shall be monitored and recorded whether the monitor is energized or not energized using oscillograph recorders of sufficient frequency response to record any change in the state of the outputs of the monitor. Monitors that have been shock tested shall not be delivered to the Government.

4.7.7.1 Definition of failure. The definition of failure of a monitor to conform to the shock requirements shall be the occurrence of any one or more of the following conditions:

(a) Breakage of any parts including mounting brackets. Minor chipping or distortion will be permitted, where such chipping or distortion cannot in any manner impair the operation of the monitor or impair the operation of the controller in which the monitor is mounted.
(b) Any relay contact chatter (change of state) of the isolation relay at any time during or after a shock blow (other than state changes caused by the normal application or removal of input power to the monitor).
(c) Any permanent change in the state of relay contacts used in the operation of indicator lights (other than state changes caused by the normal application or removal of input power to the monitor) after a shock blow.
(d) Failure of the monitor to conform to the insulation resistance test requirements of 4.7.5 conducted at the conclusion of the shock test.
(e) Failure of the monitor at the completion of the shock test to conform to the operation requirements of 4.7.3 for operation (excluding inclined tests).

4.7.8 Vibration test. Vibration tests shall be conducted in accordance with MIL-STD-167-1 to determine conformance with 3.5.2. Monitors shall be alternately energized and de-energized during the vibration testing. The state of all outputs of the monitor shall be monitored and recorded whether the monitor is energized or not energized using oscillograph recorders of sufficient frequency response to record any change in the state of the outputs of the monitor.

4.7.8.1 Definition of failure. The definition of failure of a monitor to conform to the vibration requirements shall be the occurrence of any one or more of the following conditions:

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(a) Breakage of any parts including mounting brackets.
(b) The change of state of any output of the monitor or any relay within the enclosure of the monitor at any time during the vibration test (other than state changes caused by the normal application or removal of input power to the monitor).
(c) Failure of the monitor to conform to the insulation resistance test requirements of 4.7.5 conducted at the conclusion of the vibration test.
(d) Failure of the monitor at the completion of the vibration test to conform to the operation requirements of 4.7.3 for operation (excluding inclined operation tests).

4.7.9 EMI emission and susceptibility. An EMI emission and susceptibility test shall be conducted in accordance with Parts 5 and 6 of MIL-STD-462 for CE01, CE03, CS01, RE01, RE02, and RS01 requirements. Tests shall be conducted on each input and output conductor with the monitor connected to an input voltage supply and a load. Information on conducted and radiated emission and susceptibility levels shall be recorded.

4.7.10 Ambient temperature test. Monitors shall be subjected to an ambient temperature test to determine conformance with 3.5.4. Monitors shall be energized and operated in a test chamber that shall obtain an ambient temperature of 149°F (65°C) at a rate of change between 1 to 40°F (0.5 to 22.5°C) per minute. While monitors are operating, temperatures shall be increased from an ambient temperature of 77 to 149°F (25 to 65°C) and remain at the elevated temperature until all internal components within the monitor have reached their maximum operating temperature at the elevated ambient temperature. Monitors shall be operated in accordance with 4.7.3 with the exception that no tests shall be conducted with the monitor inclined off axis. Monitors that do not pass the test criteria of 4.7.3 shall have failed the ambient temperature test.

4.7.11 Humidity test. Humidity tests shall be conducted in accordance with method 103 of MIL-STD-202. Monitors shall be exposed to humidity in accordance with test condition A. Monitors shall not be energized during the exposure period. Insulation resistance in accordance with 4.7.5 shall be measured and recorded after monitors have been conditioned for the humidity test and upon completion of the exposure period while the monitors are located in the humidity chamber.

4.7.12 Magnetic characteristic test. Monitors for nonmagnetic applications shall be tested for ferrous, eddy current and stray fields in accordance with test methods FF04, EF04 and SF04 of DOD-STD-2142.

4.7.13 Voltage spike test. Monitors shall be connected to an input voltage source that shall supply nominal input voltage at nominal frequency as specified in 3.3.1.1 for class 1 monitors and in 3.3.1.2 for class 2 monitors. Monitors shall be connected to a load that shall be adjusted such that the monitor measures a resistance simulating insulation resistance from 0.5 to 1.5 megohms. While the monitor is energized, a voltage spike with a wave shape in accordance with DOD-STD-1399, section 300 (for short-time transients) shall be applied to the input power connector from line-to-line and from line-to-ground. The magnitude

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of the voltage spike shall be 2500 volts peak for class 1 monitors and 1200 volts peak for class 2 monitors. The voltage spike shall be applied with the resistance simulating insulation resistance established as 0.5 and 1.5 megohms (two tests). After each voltage spike, the monitor shall be subjected to the test specified in 4.7.2.2 for nominal input power conditions only. The definition of failure of the voltage spike test shall be as specified in 4.7.2.3.

4.7.14 Endurance test. Monitors shall be connected to an input voltage source that shall supply rated input voltage at rated frequency as specified in 3.3.1.1 for class 1 monitors and 3.3.1.2 for class 2 monitors. Monitors shall be connected to a load that shall be varied to cause the monitor to detect a resistance simulating insulation resistance from 0.5 to 1.5 megohms. Input power shall be applied and removed from the monitor for 100,000 consecutive cycles. One cycle is defined as power applied for a minimum of 1 second and removed for a minimum of 1 second. For every 100 cycles, the value of resistance shall be set to 1.5 megohms for 80 cycles and 0.5 megohm for 20 cycles. Any mechanical damage, component failure or electrical malfunction during the endurance test shall constitute a failure.

4.8 Inspection of packaging. Sample packages and packs, and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.7.)

5.1 Packing requirements. Liquid flow meters shall be preserved level A, C or commercial, packed level A, B, C or commercial, and marked in accordance with MIL-E-17555, as specified (see 6.2.1).

(a) Navy Fire-Retardant Requirements.

(1) Treated Lumber and Plywood. Unless otherwise specified (see 6.2.1), all lumber and plywood including laminated veneer material used in shipping containers and pallet construction, members, blocking, bracing, and reinforcing shall be fire-retardant treated material conforming to MIL-L-19140 as follows:

Levels A and B - Type II - weather resistant.
Category 1 - general use.
Level C - Type I - non-weather resistant.
Category 1 - general use.

(2) Fiberboard. Unless otherwise specified (see 6.2.1), fiberboard used in the construction of class-domestic, non-weather resistant fiberboard, cleated fiberboard boxes including interior packaging forms shall meet the flame spread index and the specific optic density requirements of PPP-F-320 and amendments thereto.

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6. NOTES

6.1 Intended use. Monitors covered by this specification are intended to measure the insulation resistance of de-energized motors. Class 1 monitors are to be used with ac motors. Class 2 monitors are to be used with dc motors. Monitors are not for use with generators, with motors driving generators or with propulsion motors.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Class (see 1.2).
(c) Part number (see 1.3).
(d) Nominal input voltage (see 3.3.1.1 and 3.3.1.2).
(e) Number of remote indicators per monitor (see 3.3.2).
(f) Monitors for nonmagnetic applications (see 3.4).
(g) Level preservation and packing required (see 5.1).

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.475-1 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Technical manuals (including Technical Repair Standards (TRS) shall be acquired under separate CLINs or subCLINs). Technical manual products (i.e., manuscript copy, preliminary copy, reproducible copy, etc.) shall be developed and delivered as specified in the Technical Manual Contract Requirements (TMCR) exhibit(s) to the contract. CDRLs and DIDs shall not be used for technical manual product acquisition. Technical manual management data (plans, reports, etc.) shall be acquired under a separate data CLIN, but delivery shall continue to be specified on CDRLs and DIDs. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.	Option
3.7	Drawings, engineering and associated lists	DI-E-7031	Level 3
3.7.1	Certificate of compliance	DI-E-2121	----
4.4.4	First article inspection report	DI-T-4902	----
4.5.2	Inspection and test reports	DI-T-5329	----

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5010.12-L., AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

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6.2.2.1 The data requirements of 6.2.2 and any task in sections 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 First article. When a first article inspection is required, the item should be a first article sample. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangement for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Definitions. Definitions for electrical and electronic terms used within this specification shall be in accordance with ANSI Standard 100-84.

6.5 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract.

6.5.1 When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.6 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards which have been cleared and listed in DoD 5010.12-L (AMSDL) must be listed on a separate CDRL (DD Form 1423), included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.7 Subject term (key word) listing.

Ac or dc motor
Auxiliary equipment
Controllers, motor
Device, monitoring
Electrical equipment

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6.8 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

Preparing activity:
Navy - SH
(Project 6625-N841)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-M-24678(SH)	MONITORS, MOTOR INSULATION RESISTANCE, SHIPBOARD USE (METRIC)

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)* _____
b ADDRESS *(Street, City, State, ZIP Code)*	

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

DD FORM 1426 82 MAR

PREVIOUS EDITION IS OBSOLETE

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